SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

- **Problem Statement ID – SIH 26073**
- **Problem Statement Title- AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 155429**
- **Team Name - Astrix__**

SkyGuard AI: Real-Time Anomaly Detection in AWS

ASTRIX__

SMART INDIA HACKATHON 2026

REAL WORLD ISSUE

- Faulty sensor data
- Complex anomalies & false alarms
- Unreliable weather data
- Scalability of real-time monitoring



WHY THIS PROBLEM MATTERS

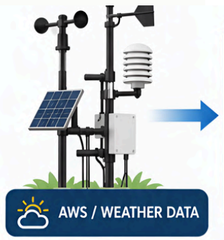
1,200+ India
AWS/Agro-AWS support weather forecasting, agriculture and disaster management.

16,300+ Global
land weather stations provide critical observations worldwide.

65.8% 2019–2024
Global surface observations increased.

→ Reliable sensor data is essential for accurate weather decisions.

PROPOSED SOLUTION



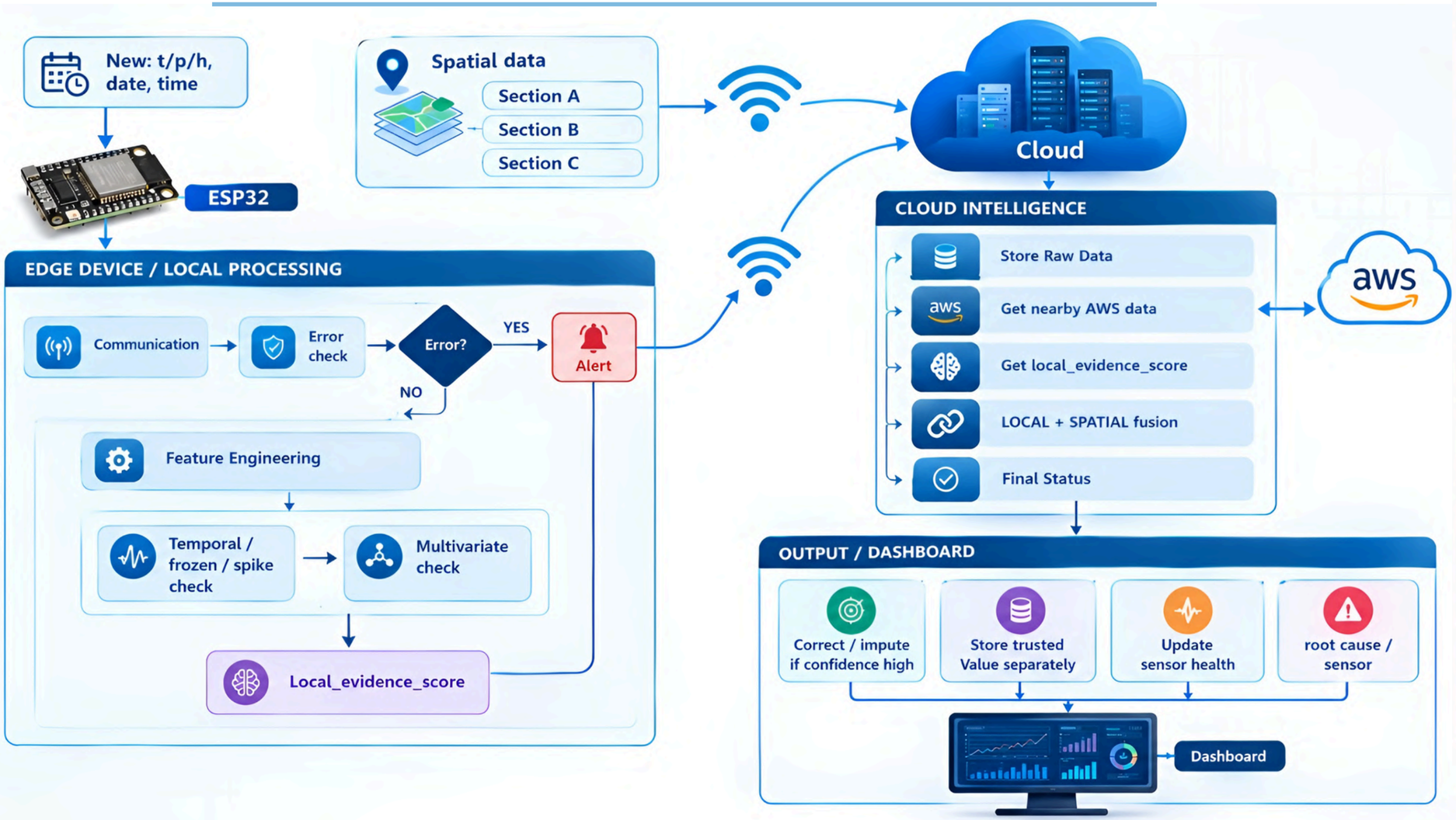
- Real time anomaly alert
- Severity & confidence score
- Root cause identification
- Sensor health status
- Explanation
- Prevent false alarm
- Trustworthy Data

Gaps in Existing Solutions	Our Proposed Solution
1 Single-variable focus	1 T-P-RH multivariate consistency
2 Weak explainability	2 Severity + confidence score + root-cause identification
3 Offline processing	3 Real-time streaming
4 Forecast–fault confusion	4 Distinguish sensor fault vs genuine weather change
5 False alarm	5 Multi-layer checks to prevent false alarms
6 Limited scalability	6 Generalize across stations
7 Limited automation & Edge deployment	7 Maintenance, correction & ESP32 deployment

Prototype Website : LINK
GitHUB Repo : LINK
Prototype Video: LINK
Youtube Presentation: LINK

TECHNICAL APPROACH

Methodology & Process of Implementation



TECHNOLOGIES USED

Edge Hardware	ESP32
Firmware	ESP-IDF, C/C++
Communication	MQTT, LoRaWAN
Cloud Ingestion	AWS IoT Core
Cloud Processing	AWS Lambda
Analytics	Python, NumPy, Pandas
Storage	TimescaleDB
Dashboard	Streamlit

ASTRIX__

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Feasibility of SkyGuard AI

01 Self-Aware
Continuously monitors sensor behavior using temporal, multivariate, and spatial checks to identify abnormal or inconsistent readings.

02 Self-Healing
Can automatically flag faulty data and, where confidence is sufficient, provide corrected or imputed values while maintaining traceability.

03 Infrastructure Reusability
Can work with existing AWS sensors and station infrastructure, minimizing hardware replacement.

04 Scalable Deployment
Its edge-cloud architecture allows additional AWS stations to be added without redesigning the complete system.

05 Cost Effective
Uses low-cost ESP32 edge hardware and lightweight local processing, while cloud resources handle larger-scale analysis.

06 Authority Integration
Provides centralized anomaly alerts, sensor-health information, and validated observations to support monitoring and maintenance.

Business Value & Potential



01 Cost & Efficiency
Reduces maintenance costs through early fault detection and better allocation of field resources.



02 Commercial Potential
Can be offered as a scalable sensor-health and weather-data monitoring service for government and private organizations.

Viability of SkyGuard AI

01 Proven Technology
Uses established ESP32, MQTT, AWS IoT Core, Python and time-series technologies.

02 Trustworthy Data
Combines temporal, multivariate and spatial checks with traceable anomaly decisions.

03 Sustainable Operation
OTA updates enable remote firmware and detection-rule updates without physical access to each station.

04 Scalable Deployment
Edge-cloud architecture allows new AWS stations to be added with minimal infrastructure changes.

05 Maintenance Support
Continuous sensor-health monitoring helps identify degradation and maintenance needs.

ASTRIX__

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Faster Response

Real-time edge detection provides early anomaly alerts.

Reduced Maintenance

Early fault detection helps reduce unnecessary inspections and downtime.

Low Complexity

Lightweight edge processing reduces dependence on continuous cloud connectivity.

Low Power

ESP32-based processing uses low computational resources, supporting power-efficient operation.

BENEFITS | IMPACT

Emergency Awareness

Reliable anomaly detection supports faster investigation of abnormal atmospheric conditions.

Environmental Impact

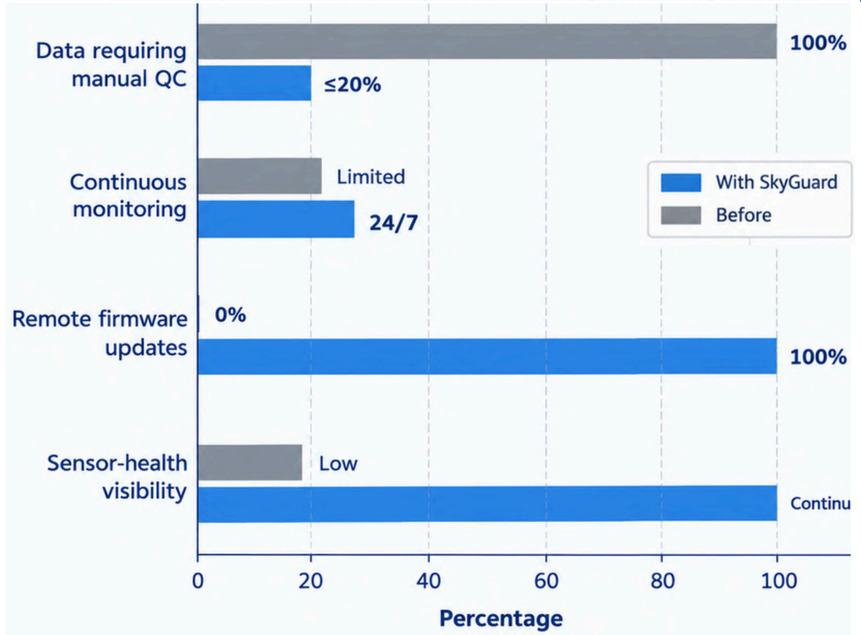
Better-quality weather data supports weather monitoring, agriculture, climate observation, and disaster management.

Economic Impact

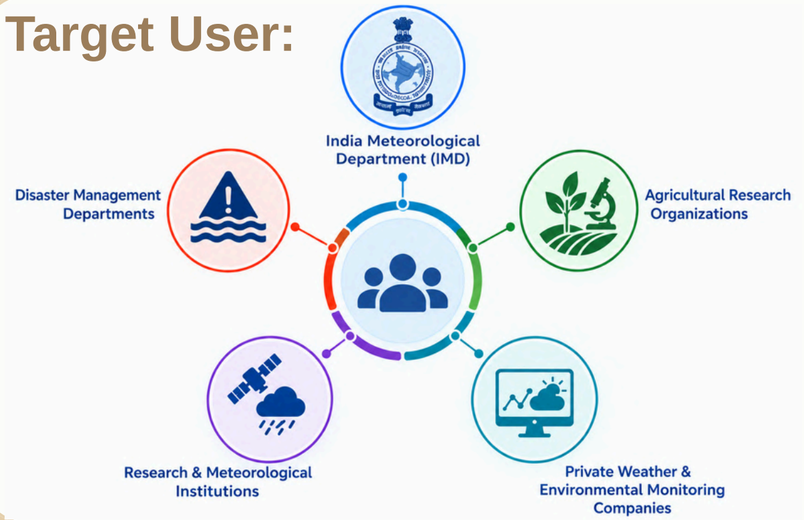
Improved data quality can reduce maintenance and data-quality costs across AWS networks.

Operational Reliability

Continuous anomaly monitoring improves the reliability of large AWS networks.



Target User:



ASTRIX__

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

regression-based analytical redundancy for sensor fault detection and localization.

LINK

Foundation for residual-based fault detection and fault isolation.

LINK

Cloud IoT connectivity & device communication

LINK

IMD MAUSAM Journal

LINK

Regression / anomaly-detection models

LINK

IMD Annual Report 2023

LINK

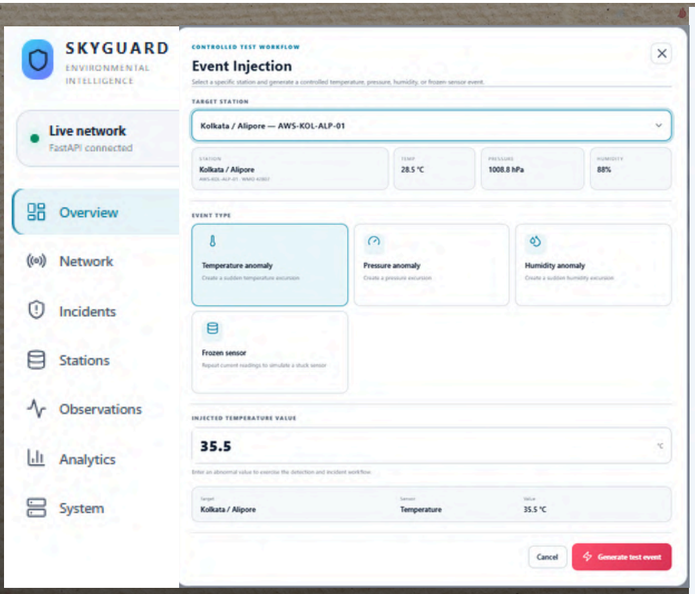
Dataset LINK

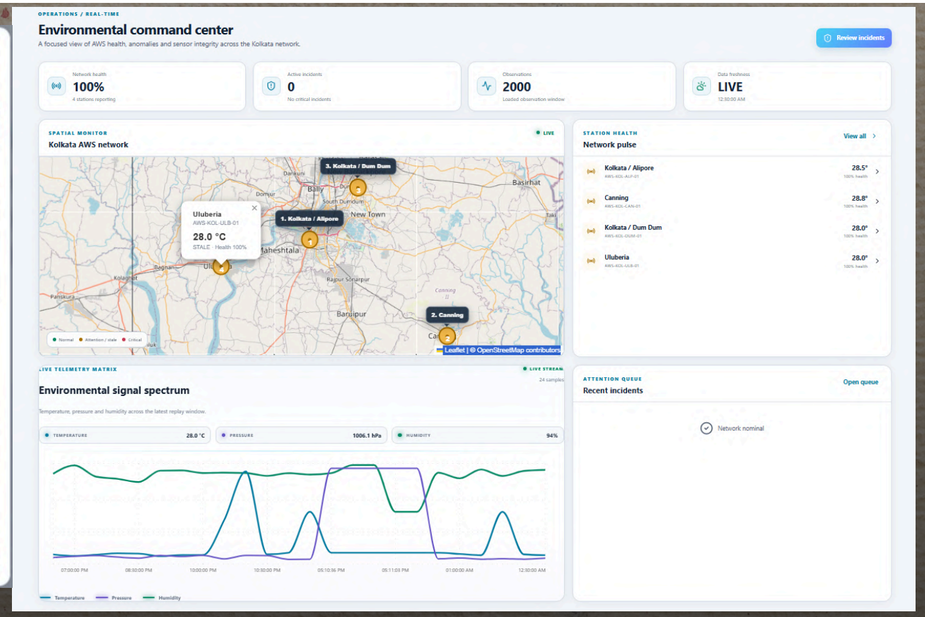
GitHub LINK

YOUTUBE

Prototype Video

PPT Presentation Video





UI / UX

